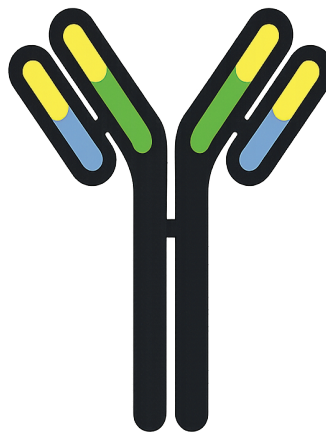
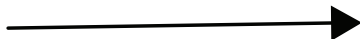




Fix L, re-estimate U



Programs in gene space

$$\mathbf{X} \approx \mathbf{L}\mathbf{F}^T$$

cells x genes

cells x GPs

GPs x genes

Programs in protein space

$$\mathbf{Y} \approx \mathbf{L}\mathbf{U}^T$$

cells x proteins

cells x GPs

GPs x proteins